

WIKIPEDIA

Bosque (programming language)

From Wikipedia, the free encyclopedia

Bosque is a free and open-source programming language designed & developed by Microsoft that was inspired by the syntax and types of TypeScript and the semantics of ML and Node/JavaScript.^{[3][4]} Design goals for the language include better software quality and improved developer productivity.^{[5][6]}

Contents

- 1 Overview
- 2 Examples
 - 2.1 Add two numbers
- 3 See also
- 4 References
- 5 Further reading
- 6 External links

Overview

Bosque was designed by Microsoft Research computer scientist Mark Marron,^[7] who describes the language as an effort to move beyond the structured programming model that became popular in the 1970s.^{[3][8]}

The structured programming paradigm, in which flow control is managed with loops, conditionals, and subroutines, became popular after a 1968 paper titled "Go To Statement Considered Harmful" by computer scientist Edsger Dijkstra. Marron believes we can do better by getting rid of sources of complexity like loops, mutable state, and reference equality. The result is Bosque, which represents a programming paradigm that Marron, in a paper he wrote, calls "regularized programming."

The Bosque specification, parser, type checker, reference interpreter, and IDE support are licensed under MIT License and available on GitHub.^[9]

Examples

Add two numbers

```
function add2(x: Int, y: Int): Int {
```

Bosque

**Bosque**

Paradigm	Multi-paradigm: functional, typed language
Designed by	Mark Marron
Developer	Microsoft
First appeared	March 3, 2019 ^[1]
License	MIT License
Filename extensions	.bsq ^[2]
Website	www.microsoft.com/en-us/research/project/bosque-programming-language/ (https://www.microsoft.com/en-us/research/project/bosque-programming-language/)
Influenced by	JavaScript, TypeScript, ML

```
    return x + y;
}

add2(2, 3)    // 5
add2(x=2, y=3) // 5
add2(y=2, 5)  // 7
```

See also



**Free and open-
source software
portal**

- Dafny
- F* (programming language)
- Free software movement

References

1. "BosqueLanguage" (<https://github.com/Microsoft/BosqueLanguage>). Microsoft. March 3, 2019. Retrieved April 29, 2019.
2. Using Bosque - Bosque Programming Language (<https://bosque-lang.com/docs/using-bosque/>)
3. Microsoft debuts Bosque – a new programming language with no loops, inspired by TypeScript (https://www.theregister.co.uk/2019/04/18/microsoft_bosque_programming_language/)
4. Bosque is Microsoft's new open source, TypeScript-inspired programming language (<https://betanews.com/2019/04/20/microsoft-bosque/>)
5. Microsoft aims for simplicity with Bosque programming language (<https://www.techworld.com.au/article/660220/microsoft-aims-simplicity-bosque-programming-language/?fp=16&fpid=1>)
6. Microsoft's New Programming Language 'Bosque' Keeps Your Code Simple (<https://fossbytes.com/microsofts-new-programming-language-bosque-keeps-your-code-simple/>)
7. Microsoft's Bosque Language Wants to Change Programming Forever (<https://medium.com/dice-insights/microsofts-bosque-language-wants-to-change-programming-forever-265d65d73c59>)
8. Microsoft Introduces Bosque, a Programming Language for Writing Easy-to-Reason-about Code (<https://www.infoq.com/news/2019/06/microsoft-bosque/>)
9. "Bosque Programming Language" (<https://github.com/Microsoft/BosqueLanguage>). *GitHub*. May 7, 2022.

Further reading

- Mark Marron (2019). "Regularized Programming with the BOSQUE Language" (https://www.microsoft.com/en-us/research/uploads/prod/2019/04/beyond_structured_report_v2.pdf) (PDF). Microsoft Research.
- Kaczmarek, Sebastian; Ibaceta, Joel (2021). *Learn Bosque Programming: Discover the world's first regularized programming language*. Packt Publishing. ISBN 978-1839211973.

External links

- Bosque Programming Language - Microsoft Research (<https://www.microsoft.com/en-us/research/project/bosque-programming-language/>)
- GitHub - microsoft/BosqueLanguage: The Bosque programming language is an experiment in regularized design for a machine assisted rapid and reliable software development lifecycle. (<https://github.com/Microsoft/BosqueLanguage>)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Bosque_(programming_language)&oldid=1220132980"

This page was last edited on 22 April 2024, at 01:03 (UTC).
Text is available under the Creative Commons Attribution-ShareAlike 4.0 License; additional terms may apply. By using this site, you agree to the Terms of Use and Privacy Policy. Wikipedia® is a registered trademark of the Wikimedia Foundation, Inc., a non-profit organization.